

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B082-Networkwalks
Date	29 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (Google Hacking Data Base) W2-PM3 (Maltego) W2-PM4 (TheHarvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network 3. Google Vulnerable Files
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers foot printing the networkwalks.com domain using multiple Kali Linux tools (W2-PM1), Google Hacking Data Base (GDHB) (W2-PM2), Maltego (W2-PM3), theHarvester (WK2-PM4) and scanning my own local network with Zenmap (W2-PM5). The first four of these modules cover the foot printing phase and the last one (PM5) covers the scanning phase, so all-together they show how an attacker moves from gathering public information to mapping live hosts on a network.

These projects were created in Week 2 of my internship at NetWorkWalks.

The commands were run on Kali Linux, Maltego, theHarvester, and on a Windows PC with Zenmap installed (scanning). Additionally, the dorks were extracted from GDHB and run on google. Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification
TheHarvester	Gathers all the public information of a specific person or organization
Maltego	Maps all publicly available information that you harvest
Google Hacking Data Base (GDHB)	Provides dork for possible vulnerable files on the internet

4. Activities Performed

4.1 Footprinting & Reconnaissance

1) Multi-Tool Recon Kali Linux

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1.2** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

I used **Curl** with the **-I** option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint **/wp-json/**.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

2) Google Hacking Data Base (GDHB)

I searched for dorks that could help me find the desired information. Using the dorks from GDHB I searched for exposed public files; it is to be noted that I did not attempt to view password protected files or camera footage.

I found 10 exposed live view cameras and 10 exposed mathematics related pdf's mostly related to universities.

3) Maltego

Maltego is essentially used to map relationships between people, domains, infrastructure, social media data, and breached information through interactive graph analysis.

I installed and ran the Maltego, Community Edition. Create a domain called networkwalks.com. Harvest an email related transform. It will then map out emails related to networkwalks.com

3) theHarvester

theHarvester is a tool that is available for download on different operating systems, I used the harvester extension on Kali Linux

It's used for gathering information of emails, sub-domains, hosts, employee names, open ports and banners from different public sources like search engines. Harvester is pre-installed on Kali Linux

- baidu as the source:

```
theHarvester -d microsoft.com -l 1000 -b baidu
```

Here **-d** is the target domain, **-l** limits the number of results to a 1000, and **-b** sets the data source. theHarvester then extracts the details and shows them on screen.

- all the sources:

```
theHarvester -d microsoft.com -l 50 -b all
```

****** all outputs and results shown in the read me file for the ease of understanding

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts. I found that it was only me on my host network so I was hoen my own IP.

The example results also included a MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

- 1. Review publicly exposed technology information**

Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

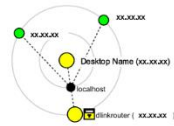
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

Zen map topology:



Harvester

[illegible]

```
(kali@kali)-[~]
$ theHarvester -d microsoft.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester *
* theHarvester 4.10.1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*****

[*] Target: microsoft.com
[*] Searching Baidu.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.
```

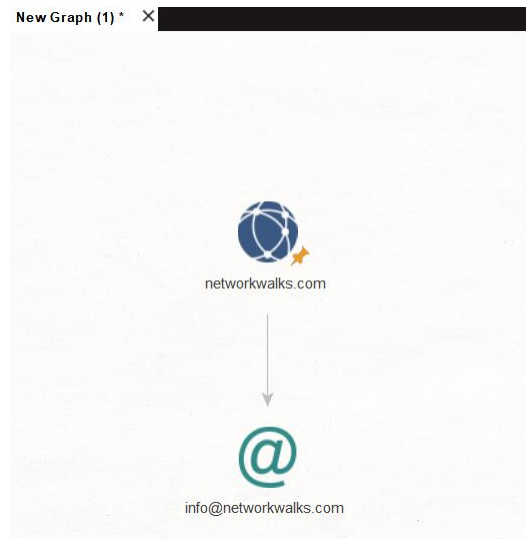
Curl -I

```
(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.ne
t" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "http
s://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/; rel="https://api.w.org"/>, <https://networkwalks.com/wp-json/wp/v2/pages/53; rel="alte
rnative"; title="JSON"; type="application/json">, <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=c12ab84d85c4d713f517f595989cbc9; path=/; domain=networkwalks.com; secure; HttpOnly
content-type: text/html; charset=UTF-8
date: Sat, 26 Sep 2020 19:40:10 GMT
server: Apache
```

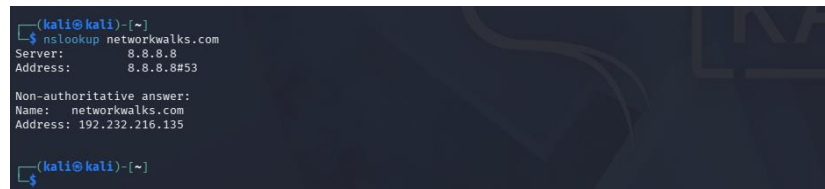
Dns recon

```
Session Actions Edit View Help
(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2020-09-29T07:14:53.971524-0400 INFO Starting enumeration for domain: networkwalks.com
2020-09-29T07:14:53.971556-0400 INFO Info: Performing General Enumeration against: networkwalks.com...
2020-09-29T07:14:54.723969-0400 INFO No answer for DNSSEC query for networkwalks.com
2020-09-29T07:14:55.446339-0400 INFO SOA ns01313.hostgator.com 50 87 144 87
2020-09-29T07:14:55.532809-0400 INFO NS ns01313.hostgator.com 50 87 144 87
2020-09-29T07:14:55.571809-0400 INFO Bind Version for 50 87 144 87: "9.10.23-BH"
2020-09-29T07:14:55.671885-0400 INFO NS ns01316.hostgator.com 192 232 236 131
2020-09-29T07:14:55.716839-0400 INFO Bind Version for 192 232 236 131: "9.10.23-BH"
2020-09-29T07:14:55.808186-0400 INFO MX mail.networkwalks.com 192 232 236 135
2020-09-29T07:14:58.828581-0400 INFO A networkwalks.com 192 232 236 135
2020-09-29T07:15:02.434319-0400 INFO TXT networkwalks.com v=spf1 -a -mx -ip4:50.87.166.87 -include:websitelwelcome.com -all
2020-09-29T07:15:02.436916-0400 INFO TXT networkwalks.com google-site-verification=rr04emqg0wY3Xenn120Wkq75K-Ij-mjieg-usY
1
2020-09-29T07:15:03.204983-0400 INFO Enumerating SRV Records
2020-09-29T07:15:06.678807-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 44
3
2020-09-29T07:15:06.679208-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 44
3
2020-09-29T07:15:06.679463-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 44
3
2020-09-29T07:15:06.679757-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2020-09-29T07:15:06.679914-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 44
3
2020-09-29T07:15:06.680267-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 44
3
2020-09-29T07:15:06.680362-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 44
3
2020-09-29T07:15:06.680499-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2020-09-29T07:15:06.683150-0400 INFO 8 Records Found
2020-09-29T07:15:06.683552-0400 INFO Completed enumeration for domain: networkwalks.com
```

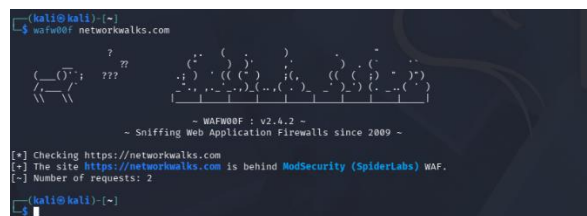
Maltego

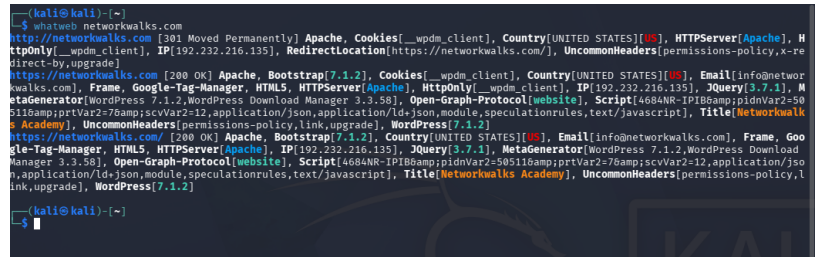
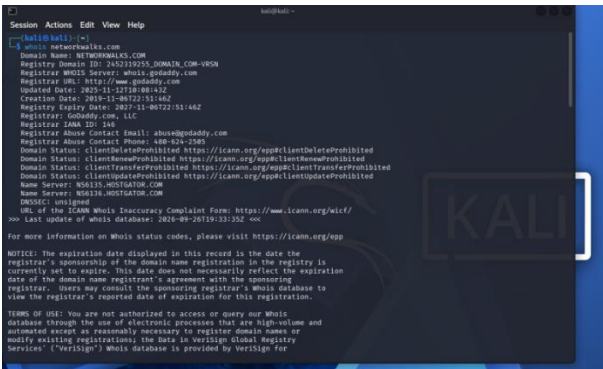
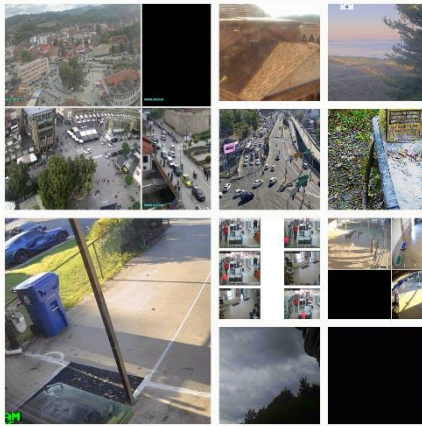


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Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub